

SMASH - WALL COATING ("LỚP PHỦ") SPEC

Difficulty ramp across Mission 1-3, and how to author it in map JSON

Companion to SMASH_MapJSON_Spec_and_Review.txt and
SMASH_JSON_Cloning_Handbook.txt

Project: Demolition Contracts (FierceTigerGameJam) Revision 2 - 2026-08-30
(section 5 implemented)

WHO THIS IS FOR

The AI or person producing map JSON, and whoever tunes the engine. Section 1 is the ground truth about what "coating" actually is - read it before acting on sections 2-3, because the feature does not work the way the name suggests.

DECISION ON RECORD

OPTION A is the priority. OPTION B is the fallback, to be used only if A cannot be delivered. Both are written out in full in section 3 and 4 so the choice can be made without going back to the conversation.

1. WHAT "COATING" IS - GROUND TRUTH

1.1 IT IS NOT A MATERIAL, A MESH, OR AN EXTRA SURFACE LAYER

There is a Wall Panel mesh (FBX/Brick_Wall.fbx, FBX/Concrete_Wall.fbx) that the engine stretches across an intact wall instead of welding cubes. It is PURELY COSMETIC. KnockdownLayoutMapAuthoring.ResolveWallDurability() computes hit points from the blocks, and it runs whichever mesh path was taken. The panel adds no hit points, no thickness, no extra shot. Do not design difficulty around the panel. It is a rendering detail.

1.2 THE REAL MECHANIC IS THE WALL ITSELF

A group of cells sharing a wall_id becomes ONE rigidbody whose hit points are the sum of its blocks. Two consequences, and the second is the one that matters here:

- a) Pooled hit points. A 20-cell brick wall is 60 HP in one body; the same 20 cells loose are twenty separate 3 HP targets.
- b) A SEPARATE DAMAGE NUMBER. BulletDefinition.MaterialDamage carries both blockDamage and wallDamage per material per bullet level. The engine picks by target: a wall takes wallDamage, a loose block takes blockDamage. The tooltip in BulletDefinition.cs states the intent exactly: "Lower than blockDamage is what makes a shot chip a wall while destroying a lone block."

So: CELLS IN A NAMED WALL = COATED. LOOSE CELLS = UNCOATED.

1.3 THE NUMBERS AS SHIPPED (Assets/GameJam/Config/Bullets/)

ammo	material	blockDamage	wallDamage	shell factor
Rock Lv1	glass	5	5	none
	brick	3	1	3.0x
	concrete	0	0	impossible
Rock Lv2	glass	6	6	none
	brick	4	3	1.3x
	concrete	0	0	impossible
Cannon Lv1	glass	8	8	none
	brick	6	5	1.2x
	concrete	4	2	2.0x
Cannon Lv2	glass	10	10	none
	brick	8	7	1.1x
	concrete	7	5	1.4x

1.4 THREE FACTS THAT CONSTRAIN EVERY DESIGN BELOW

- GLASS HAS NO SHELL AT ALL. wallDamage == blockDamage at every level of every ammunition. Coating glass changes difficulty by exactly zero. Grouping glass into walls is still worth doing, but as an ASMR knob: a glass wall releases every pane in one second instead of popping one by one.
- CONCRETE IS IMMUNE TO ROCK. Zero, not slow. Any level whose clear condition depends on breaking concrete is gated behind owning the Cannon.
- THE SHELL IS STRONGEST EARLY AND FADES. Rock Lv1 vs brick is a 3x shell; by Cannon Lv2 it is 1.1x. Coating is therefore a much bigger difficulty lever for a player on starter gear than for an upgraded one.

2. THE OBSTACLE - WHY A LITERAL READING OF OPTION A BREAKS

"Uncoated" means loose cells, and every loose cell is its own rigidbody on the opening frame. The budget in the spec is 150 loose bodies; beyond that Android frame time collapses. Measured against the maps as they stand today:

LV	map	entries	brick	glass	concrete
1	map_001_hollow_brick_hut	77	96	-	-
2	map_002_brick_courtyard_tower	104	126	-	-
3	map_003_three_storey_brick_house	217	283	-	-
4	map_004_glass_belt_warehouse	166	138	60	-
5	map_005_two_storey_glass_house	184	204	26	-
6	map_006_large_glass_school	222	222	48	-
7	map_007_concrete_frame_school	220	138	28	84
8	test_03 (Village Gatehouse)	354	92	56	216
9	map_009_hollow_concrete_tower	268	218	18	76

(brick/glass/concrete columns are CELL counts, not entries)

Mission 1 under Option A wants brick uncoated on LV1-6. Loose body count that would produce: LV1 96, LV2 126, LV3 283, LV4 198, LV5 230, LV6 270. Everything from LV3 up blows the 150 cap, some by nearly double. The same problem hits Mission 1 LV8, whose 216 concrete cells would all be loose.

This is not a reason to abandon Option A. It is a reason to implement it the way section 5 describes rather than by literally un-grouping cells.

3. OPTION A - THE PRIORITY DESIGN

Read "coating on material M" as: cells of material M take wallDamage instead of blockDamage, i.e. they behave as an armoured shell until the shell is broken.

MISSION 1 - residential / school

- LV 1-6 no coating on anything. Brick comes off block by block; this is where the player learns that bricks are individually cheap.
- LV 7-9 BRICK COATED. First contact with a shell. On Rock Lv1 this is a 3x jump, which is exactly the intended difficulty step. Concrete (LV7-9) stays uncoated.
- Glass never coated anywhere in Mission 1 (no effect anyway); still grouped into walls per pane group for the shatter moment.

MISSION 2 - beach architecture

- LV 1-9 BRICK COATED throughout. The brick shell is now the baseline.
- LV 7-9 CONCRETE COATED as well. Second shell introduced, and on Cannon Lv1 concrete is a 2x shell - the hardest single step in the campaign so far.
- Glass grouped for shatter, never counted as coating.

MISSION 3 - space / spaceport

- LV 1-9 BOTH BRICK AND CONCRETE COATED from level 1. No ramp-in; the player is expected to arrive with upgraded ammunition.
- LV 9 BOSS. Designed to be very hard to solve, not merely tanky. See the design note below - raw hit points is the wrong way to do

this.

THE SHAPE OF THE RAMP, STATED PLAINLY

M1 LV1-6	0 shells	
M1 LV7-9	1 shell	(brick)
M2 LV1-6	1 shell	(brick)
M2 LV7-9	2 shells	(brick + concrete)
M3 LV1-9	2 shells	(brick + concrete)

LEVEL 9 OF MISSION 3 - HOW TO MAKE IT HARD HONESTLY

Do not reach for more hit points. A wall of 400 concrete cells is not difficult, it is long. Difficulty comes from the shot economy:

- one load-bearing element that must be found (a hidden pillar behind a glass belt, reachable only through a window)
- a collapse that only completes if the order is right (fell the skybridge before the tower it leans on, and the tower survives)
- a concrete core that cannot be shot directly, only crushed by dropping something heavy on it
- ammunition budget tuned so a wasted shot is felt

Sections S1-S5 of the Blueprint document cover these patterns.

4. OPTION B - THE FALLBACK

Leave the current behaviour exactly as it is: every structural cell is grouped into a named wall by the face/storey algorithm, so everything is effectively coated everywhere, and difficulty comes only from entry count, material mix and ammunition budget.

Use this only if section 5's implementation cannot be delivered. It is not wrong - it is simply the state the campaign already ships in, and it wastes the coating as a progression lever.

5. HOW TO IMPLEMENT OPTION A - RECOMMENDED

5.1 THE PROBLEM IN ONE SENTENCE

Coating is currently welded to grouping: to remove a shell you must un-group cells, and un-grouping costs rigidbodies the platform cannot pay for.

5.2 THE FIX - DECOUPLE THEM WITH ONE FLAG

Let a wall stay a wall for physics and rendering, but carry a flag that says whether it is armoured. Unarmoured walls take blockDamage.

JSON - the flag is an OPT-OUT, absent means armoured:

```
"wall": { "wall_id": "front_brick_b0_3", "bare": true }
```

It is written as an opt-out rather than "armored": false on purpose. JsonUtility gives a field the JSON does not mention the type default, and for bool that is false; an "armored" field would have depended on field initialisers running and would have silently disarmed every existing wall if that assumption were ever wrong. "bare" cannot fail that way.

IMPLEMENTED 2026-08-30 - four edits, already in the branch:

- | | |
|-------------------------------------|---|
| 1. KnockdownMapDefinition.cs | - KnockdownMapWallRef.bare, plus KnockdownMapBlock.WallArmored. |
| 2. KnockdownLayoutMapAuthoring.cs | - WallBuild.Armored, set in CreateWallBuild as the AND over the group's blocks, passed to Initialize. |
| 3. BreakableWall.cs | - serialized "armored" field (so a baked map prefab keeps it), IsArmored, Initialize(..., bool isArmored = true). |
| 4. GridKnockdownCannonProjectile.cs | - ResolveDamage is now called with (wall != null && wall.IsArmored), so |

a bare wall returns blockDamage. No new damage maths.

What this buys:

- Option A becomes exact, per material, per level, with no compromise.
- Loose body count never rises; walls stay grouped, frame time unchanged.
- Every existing map keeps behaving identically (absent flag = armoured).
- The knob is per wall, not per map, so a single building can mix an armoured concrete frame with unarmoured brick infill - which is the A7 exoskeleton pattern from the Handbook, finally readable in play.

5.3 AUTHORIZING RULE ONCE THE FLAG EXISTS

Set "bare": true on every wall whose material is uncoated at that level per section 3. Leave it out otherwise. Nothing else about wall grouping, sizes, footing or brick_2x1 merging changes - keep following the Handbook.

6. FALLBACK IMPLEMENTATION - IF THE ENGINE CANNOT CHANGE

If the flag in section 5 is not available, approximate Option A with grouping, under a hard rule so the result stays shippable:

- R1 LOOSE CAP IS ABSOLUTE. Never exceed 150 loose bodies on the opening frame, whatever the design says. A level that stutters is worse than a level with the wrong shell.
- R2 WHEN UNCOATED CELLS EXCEED THE CAP, WALL THE STRUCTURE, LOOSEN THE SKIN. Keep in walls: foundations (row 0), corner piers, lintel bands, roof and floor slabs - the cells that hold the building up. Leave loose: the outer face panels between them. The player then experiences shooting bricks off one at a time, which is the feel of "no coating", while the skeleton stays cheap and the building still collapses properly.
- R3 NEVER LEAVE A BODY UNFOOTED. The footing repair pass still applies; a loosened skin cell that ends up floating hangs forever and blocks the 100% clear.
- R4 SHRINK BEFORE YOU BREAK THE CAP. A level that cannot be authored uncoated within 150 loose bodies should lose cells - more openings, hollower core - rather than gain walls. Holes are free budget.
- R5 ACCEPT THE APPROXIMATION AND WRITE IT DOWN. Deliver the manifest line with a "coating: partial (n% loose)" note so the tuning pass knows what it is looking at.

7. CHECKLIST FOR EACH DELIVERED MAP

- [] coating per material matches section 3 for that mission and level
- [] glass never counted as coating; still grouped per pane group
- [] if using section 5: "bare": true present on exactly the right walls
- [] if using section 6: loose bodies <= 150, structure walled, skin loose
- [] every body footed; validator reports 0 errors
- [] entries within the level band; wall sizes 4-24 cells
- [] manifest line states: entries, walls, loose, coating per material
- [] concrete-bearing levels flagged as Cannon-gated (Rock does 0 to concrete)

END